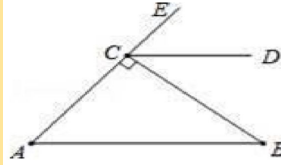


Problem:

Description: “ $\angle ABC = 40^\circ$, $CD \parallel AB$,
BC is perpendicular to AC.
Find the measure of $\angle ECD$ ”



Construction:

Shape(CA, AB, BC), Shape(EC, CD) ...

Construction (extended):

Shape(AB, BC, CA), Shape(BC, CA, AB) ...

Conditions:

Equal(MeasureOfAngle(ABC), 40),

ParallelBetweenLine(CD, AB) ...

Goal:

Value(MeasureOfAngle(ECD))

Answer: 50

Proof:

1) *parallel_property_alternate_interior_angle(1, CD, AB)*

2) *angle_addition(1, ECD, DCB)*

3) *adjacent_complementary_angle(1, ECB, BCA)*

Theorem Dictionary (GDL):

parallel_property_alternate_interior_angle(AB, CD):

{

1: {

premises:

ParallelBetweenLine(AB, CD) & Line(AD),

conclusions: *Equal(MeasureOfAngle(BAD), ...)*

},

2: { ... },

...

},

angle_addition(ABC, CBD): {

1: { ... },

...

},

...